

Understanding Analysis, 2nd Edition

Chapter 5 - Section 4.6 - Sets of Discontinuity

1. Define $D_f \subset \mathbf{R}$ to be the set of points where the function f is discontinuous. Using modifications of Dirichlet / Thomae functions, construct a function $f : \mathbf{R} \rightarrow \mathbf{R}$ so that
 - (a) $D_f = \mathbf{Z}^c$.
 - (b) $D_f = \{x : 0 < x \leq 1\}$.

Solution:

- (a) Let $D(x)$ be the distance from x to the integer closest to it. That is, $D(x) = \min(|x - \lfloor x \rfloor|, |x - \lceil x \rceil|)$.

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is an integer or irrational} \\ D(x) & \text{if } x \text{ is rational} \end{cases}$$

It's continuous at integer x . If $(x_n) \rightarrow x$ then $f(x_n) \rightarrow 0$ for both rationals and irrationals in (x_n) .

It's discontinuous at rational x . Take a sequence of irrationals $(x_n) \rightarrow x$. We have $f(x_n) \rightarrow 0$ but $f(x) > 0$.

It's discontinuous at irrational x . Take a sequence of rationals $(x_n) \rightarrow x$. We have $f(x_n) \rightarrow D(x) > 0$ but $f(x) = 0$.

- (b) Rename the function in part(a) to g . We'll reuse it here. Let

$$f(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ g(x) & \text{if } 0 < x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

It's continuous on $x < 0$ because it's constant there.

It's continuous at 0 because by part (a) $x \rightarrow 0 \implies g(x) \rightarrow 0$ and $g(0) = 0$.

It's discontinuous at $0 < x < 1$ by part(a).

It's discontinuous at 1 because by part(a), $x^- \rightarrow 1 \implies g(x) \rightarrow 0$ but $g(1) = 1$.

It's continuous on $x > 1$ because it's constant there.

2. Given a countable set $A = \{a_1, a_2, a_3, \dots\}$, define $f(a_n) = 1/n$ and $f(x) = 0$ for all $x \notin A$. Find D_f .

Solution: It's not too difficult to visualize that f must be discontinuous at $c \in A$. Any δ -neighborhood $V_\delta(c)$ must contain an $x \notin A$ (if not then there exists an interval that consists only of $x \in A$, making A uncountable). Then any $V_\delta(c)$ contains an x such that $f(x) = 0$ and since $f(c) > 0$, it's discontinuous.

What about $c \notin A$? If $A = \mathbf{Z}$ then it's clearly continuous. If $A = \mathbf{Q}$ it's less obvious, but turns out it's still continuous.

Given $\epsilon > 0$, let $N > \frac{1}{\epsilon}$. Since N is bounded, we can take $m = \min_{1 \leq i \leq N} |a_i - c|$. Then for $\delta < m$, $V_\delta(c)$ contains only those x where $f(x) = 0$ if $x \notin A$ or $f(x) < \frac{1}{N} < \epsilon$ if $x \in A$. So f is continuous at $c \notin A$.

In conclusion, $D_f = A$.